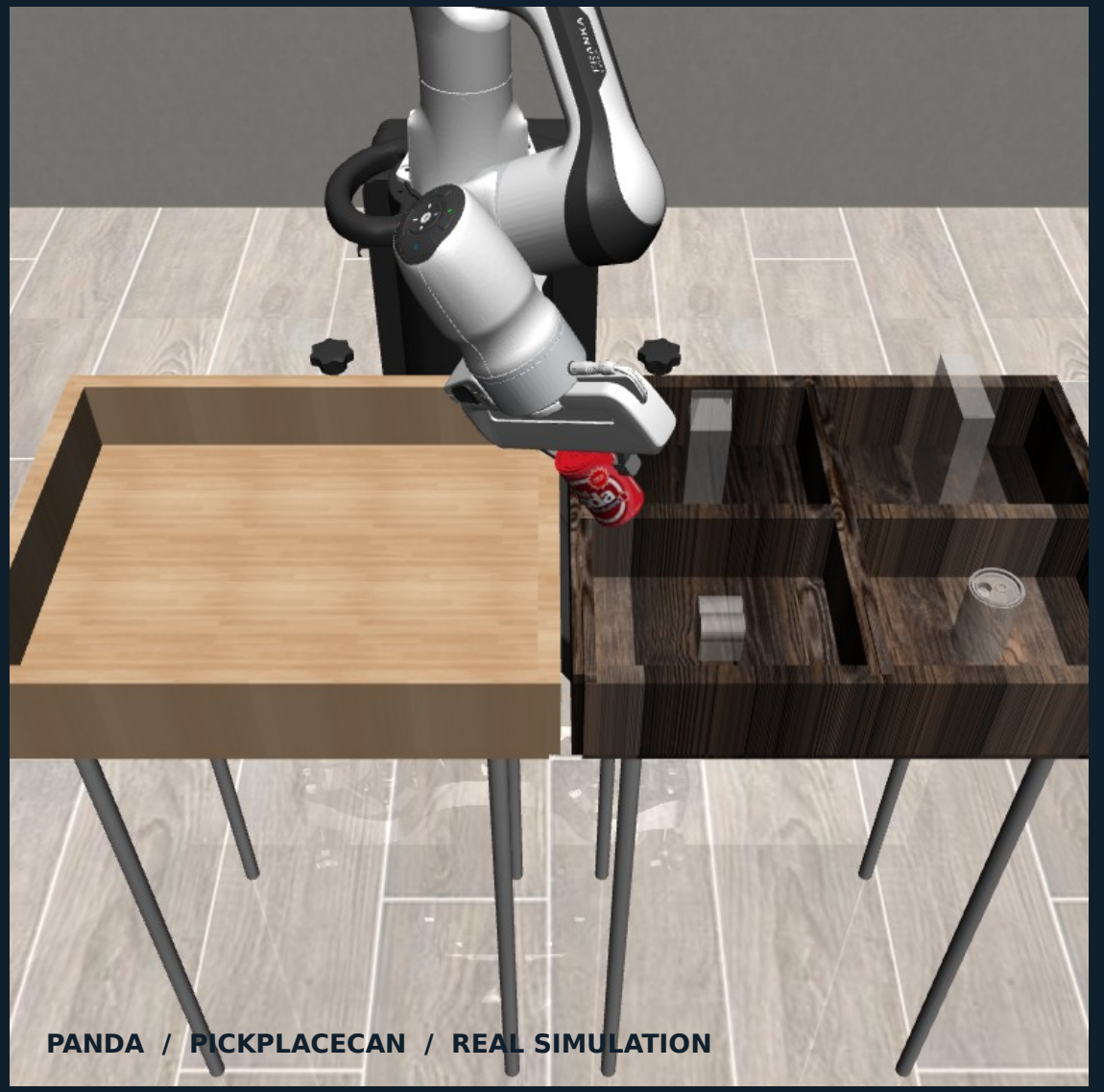
Manipulation Learning Lab

From Demonstrations to Robot Policies

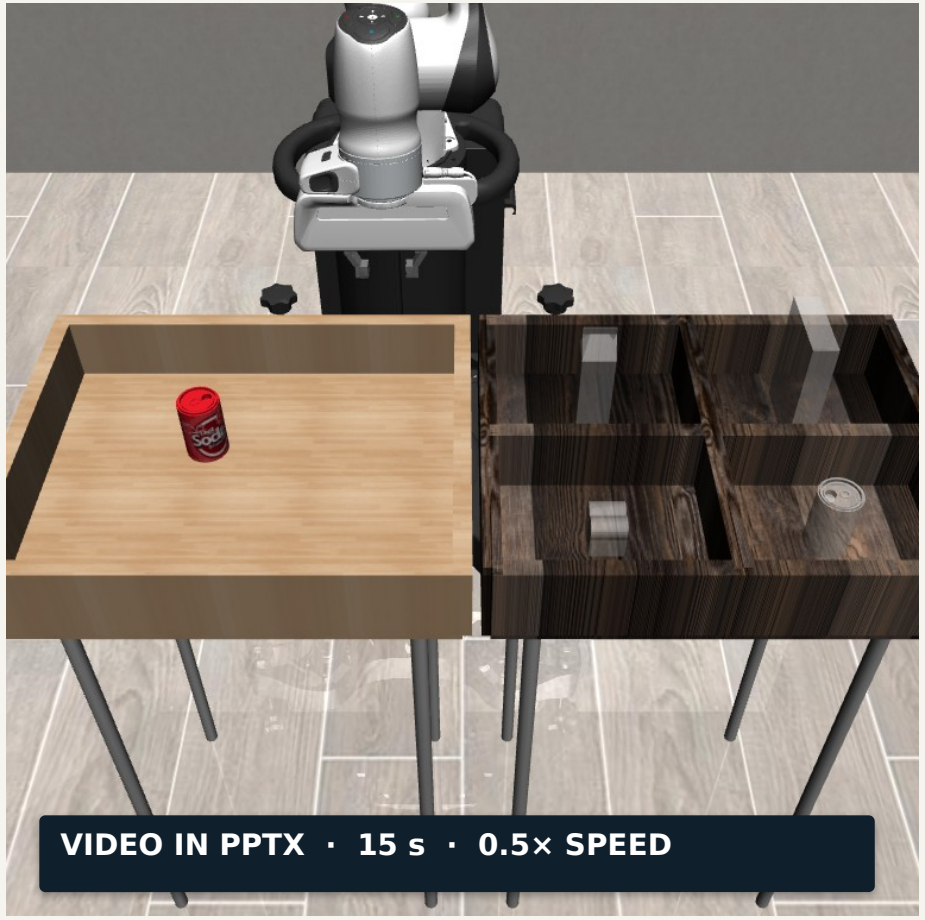
Learning Robot Manipulation from Demonstrations

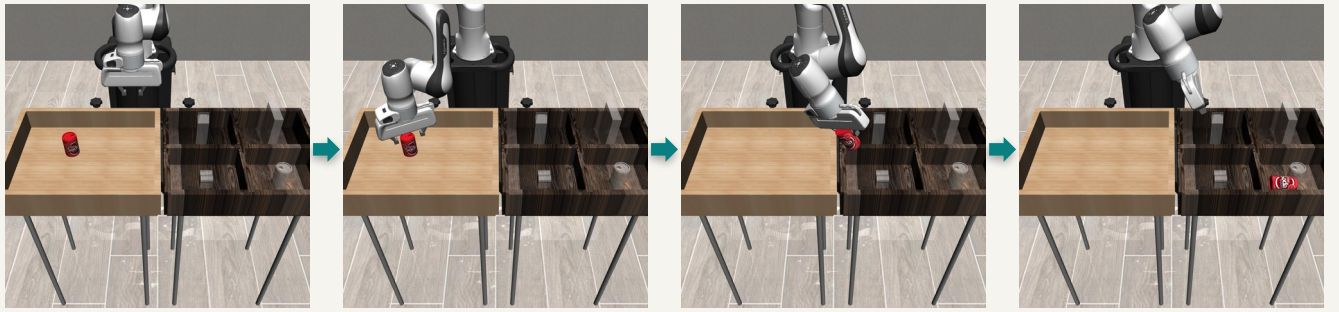
01 Watch | 02 Learn | 03 Try



What Will the Robot Do?

Can a robot learn to pick and place objects?





Reach **Grasp** **Transport** **Place**

ROBOT

Panda arm

TARGET OBJECT

Red can

GOAL REGION

Can-marked bin

Success = inside + released + stable

Target margin · open gripper · no contact

Stable for 1 s: displacement ≤1 cm; speed ≤5 cm/s

Demonstration → Training → Inference

01

Demonstration

A person shows the task.

Record current state and the next action.

02

Training

Learn a state → action rule.

Compare with the human action.

Update the model weights.

03

Inference

Let the policy act.

Observe → act → observe again.

Repeat 20 times per second.

200 HUMAN DEMONSTRATIONS · SPLIT BY COMPLETE EPISODE



Validation selects saved weights. Test results never select the model.

What Does the Policy See and Do?

OBSERVATION / CURRENT STATE

16 numbers

3 + 4	Gripper-tip position + orientation
2	Two finger joint positions
3 + 4	Can position + orientation

World frame · Z points up

Positions in meters

POLICY

A learned function

ACTION / NEXT COMMAND

7 numbers

3	Move in X, Y, Z
3	Rotate the gripper
1	Close (+) or open (−)

Commands normalized to [−1, 1]

Per axis: ±0.05 m; ±0.5 rad (axis-angle)

The policy reads state values; the videos help YOU see what happens.

Behavior Cloning: Learn the Next Action

Predict the human action. Measure the error. Improve the prediction.

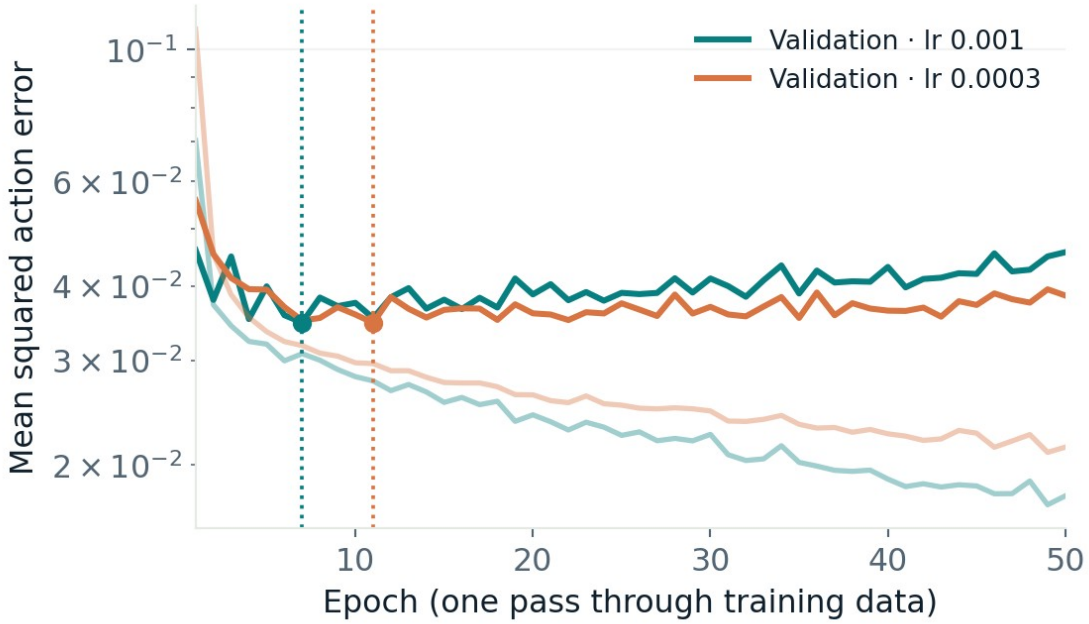
THE STUDENT MODEL



Three hidden layers, 256 units each
A feedforward neural network (MLP)

Loss = mean squared difference between predicted and human actions

Learn from training data.
Select saved weights using validation loss.



Real example · pale lines: training
Dotted lines: validation-selected epochs (7 and 11)

Your Assignment: Four Notebooks

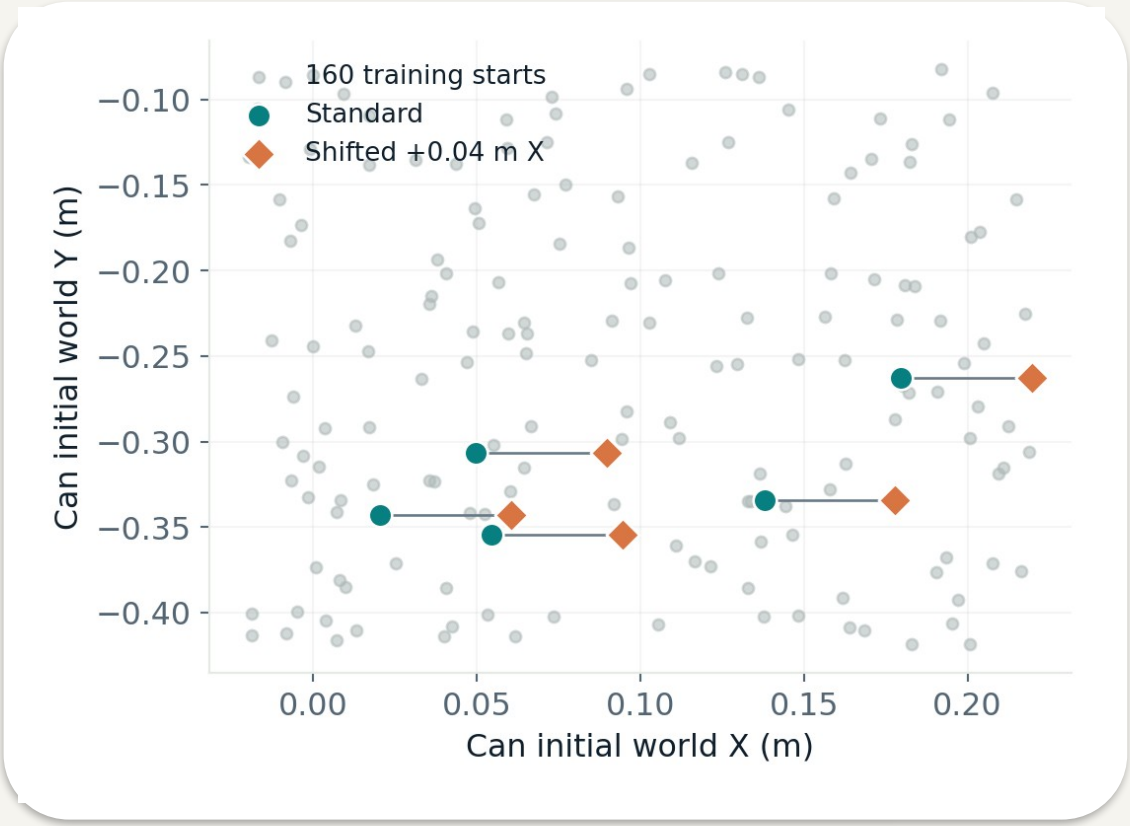
Use the prepared course server. Open 01 → 02 → 03 → 04 in Jupyter.

01	**Explore demonstrations** Play a human demo; inspect inputs and actions.	20 min	Q1
02	**Train two BC policies** Run both rates; save models, curves and evaluations.	50–60 min	Q2-Q3
03	**Compare evaluation** Read the SAME saved cases; watch outcomes.	40–50 min	Q4
04	**Analyze a real failure** Select ≥3 steps; mark the first error; export evidence.	20–30 min	Q5

ONLY CHANGE THE LEARNING RATE 0.001 vs 0.0003 · Same data, model, seed 17, 50 epochs, batch 256 & device

Evaluate Behavior, Not Just Loss

2 models × (5 standard + 5 shifted starts) = 20 rollouts



ONE COMPLETED EXAMPLE / STABLE PLACEMENTS

Learning rate	Standard	Shifted
0.001	2 / 5	3 / 5
0.0003	5 / 5	5 / 5

Same source episode + same seed
Shift only the can: +0.04 m along world X

Five cases are a coarse estimate. A position shift does not prove out-of-distribution testing.

Make a Failure Explainable

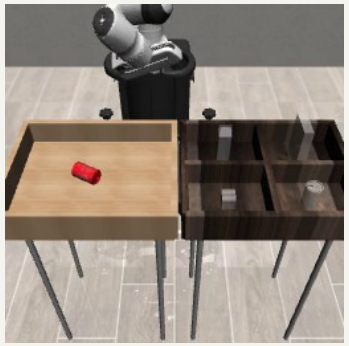
Find the first observable error. Separate what you see from why it happened.



Step 1

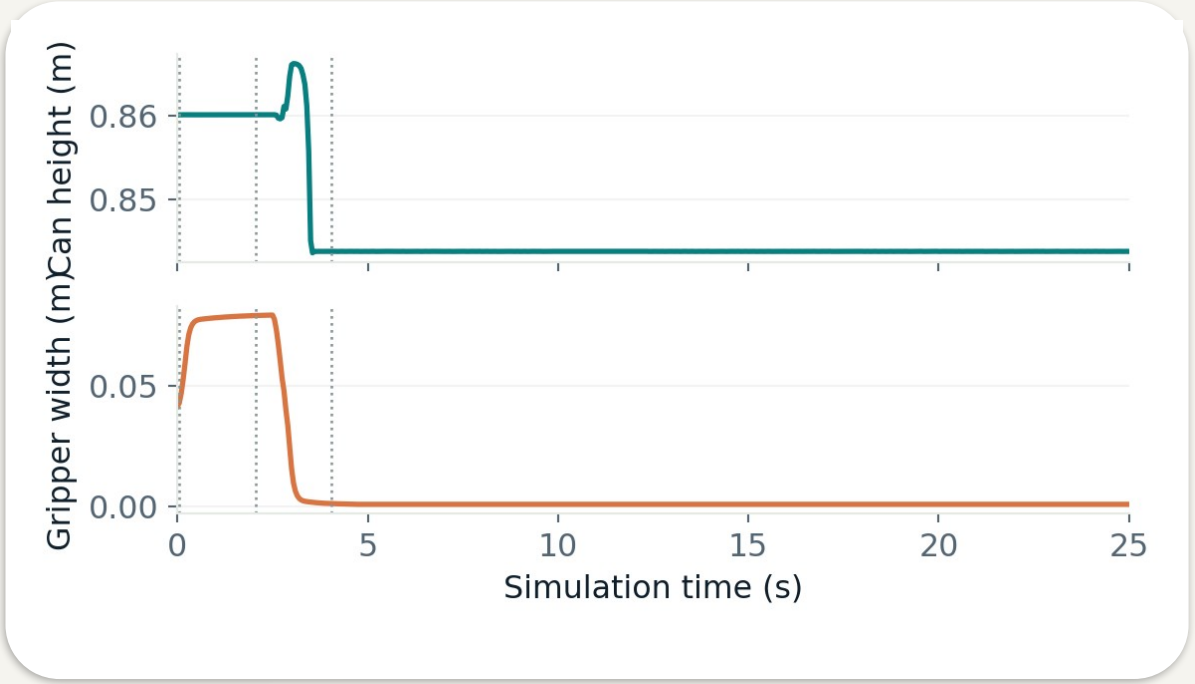


Step 41



Step 81

One real failure · primary student model
At step 81, the can remains in the source bin.



01 Play video

02 Select steps

03 Mark error

Observation → hypothesis → a test

Notebook 04 links selected frames to logged state. Check the recorded-frame offset.

What to Submit—and What Earns Credit

YOUR GROUP SUBMITS

1–2 page PDF + evidence ZIP

Answer Q1–Q5 in your own words.
Download the ZIP from Notebook 04.
Keep both models and all outcomes.

Generated evidence is an appendix.
Blank answers or no error marker = DRAFT.

100 POINTS / EVIDENCE & REASONING

20	Demonstration understanding
25	Training & controlled experiment
30	Policy evaluation
25	Failure analysis & discussion

A well-explained failure can earn full credit.